WEEK 1 · LESSON 1 OF 24 · HANDOUT

# Defining AI: AI, Machine Learning and Deep Learning

Study time: **about 40 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=4RixMPF4xis — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50: Artificial Intelligence — full lecture — https://www.youtube.com/watch?v=eey91kzfOZs

**Your goal in this lesson:** Give a precise definition of artificial intelligence, explain how AI, machine learning and deep learning fit inside each other, and place a real product in the right circle.

## A precise definition

In everyday talk, "AI" can mean almost anything. In this course we use the definition most university textbooks use:

> ***Artificial intelligence*** *is the study and building of systems that* ***perceive*** *their surroundings,* ***reason*** *about what they perceive, and* ***act*** *to reach a goal — doing tasks that normally need human intelligence.*

Notice what the definition does *not* say. It does not say "a computer that thinks like a human". A system is intelligent because it reaches goals well, not because it copies the human brain. An aeroplane flies very well without flapping its wings like a bird.

## Three circles inside each other

Picture three circles, one inside the other:

| Circle | What it means | Example |
|---|---|---|
| **Artificial intelligence** (biggest) | Any technique that makes machines act intelligently — including hand-written rules and search | A chess program that searches millions of moves |
| **Machine learning** (inside AI) | Systems that **learn patterns from data** instead of following only hand-written rules | A model that predicts loan risk from past loans |
| **Deep learning** (inside ML) | Machine learning with **many-layered neural networks**, trained on large data | Face unlock on your phone, ChatGPT |

So every deep-learning system is a machine-learning system, and every machine-learning system is an AI system — but not the other way round.

## Two big families of AI

- **Symbolic AI (rule-based)** — people write the knowledge as rules and logic: *IF fever AND rash THEN check for measles.* Strong when rules are clear; weak when the world is messy.
- **Statistical AI (learning-based)** — the system learns from examples. Strong on messy data like photos, speech and language; weaker at explaining itself.

Most modern products use learning, but many still mix both. A bank's fraud system may use a learned model *and* hard rules such as "block any transfer above the daily limit".

## Narrow AI and general AI

- **Narrow AI** does one kind of task: translating, recognising faces, recommending videos. **All AI that exists today is narrow**, even very capable chatbots.
- **Artificial general intelligence (AGI)** would match people across almost all thinking tasks. It is a research goal and a topic of debate, not a product you can buy.

Be careful with headlines. "AI can now do X" usually means *one narrow system* does X under certain conditions.

## Key terms

- **Agent** — anything that perceives and acts (you will study agents in Lesson 3).
- **Model** — the learned pattern stored after training.
- **Training** — the process of learning a model from data.
- **Inference** — using a trained model to make a prediction on new input.
- **Algorithm** — a step-by-step method; a learning algorithm produces a model.

**Remember this:** AI is the big field; machine learning is the part that learns from data; deep learning is the part of ML that uses large neural networks. All of today's AI is narrow.

Name: _____ Date: _____

## Activity

### Task 1

List five apps you used this week (for example MoMo, Google Maps, WhatsApp, YouTube, Irembo).

### Task 2

For each one, name one feature that might use AI.

### Task 3

Place each feature in a circle: rule-based AI, machine learning, or deep learning. Write one sentence explaining your choice.

### Task 4

Keep this list — you will use it again in the Week 1 project.

## Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Which definition of AI matches this lesson best?

   A. Systems that perceive, reason and act to reach goals that normally need human intelligence
   B. A computer that has feelings
   C. A robot with a human body
   D. Any program that uses the internet

2. How are AI, machine learning and deep learning related?

   A. Deep learning contains machine learning, which contains AI
   B. Machine learning is bigger than AI
   C. They are three unrelated fields
   D. AI contains machine learning, which contains deep learning

3. A system follows the rule 'IF fever AND rash THEN check for measles'. What kind of AI is this?

   A. Symbolic (rule-based) AI
   B. Unsupervised learning
   C. Deep learning
   D. General AI

4. Which statement about today's AI is correct?

   A. All AI that exists today is narrow AI
   B. Most AI products are general AI
   C. Narrow AI cannot be useful
   D. AGI is sold in app stores

5. What is 'inference'?

   A. Using a trained model to make a prediction on new input
   B. Collecting data
   C. Writing rules by hand
   D. Deleting a model

6. Face unlock on a smartphone is best described as…

   A. Deep learning
   B. General AI
   C. Not AI at all
   D. A hand-written rule

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